Mingyu Li

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Education

2022–present **PhD in Astronomy**, *Department of Astronomy, Tsinghua University*

2018–2022 **Bachelor of Physics**, *Department of Physics, Tsinghua University*

Thesis: Circumgalactic Lyman-alpha Nebulae with Unknown Powering Mechanisms (Advisor: Zheng Cai)

Minor in Statistics, *Center for Statistical Science, Department of Industrial Engineering, Tsinghua University*

Research Interests

High-z galaxy
- Chasing first stars (Population III) with JWST observations
- Metal enrichment in the early Universe

Baryon cycle
- Galaxy baryon cycle through ISM/CGM emission observations
- Circumgalactic nebulae, cold gas accretion, and cosmic web

Galaxy cluster
- Galaxy overdensity, group, and (proto)cluster
- Environmental effects on galaxy evolution

Awards & Prizes

2025 **National Scholarship**[1]

2022-2024 **Tsinghua-Ubiquant Scholarship**[2] (2 Years)

Undergraduate

2021 **National Astronomical Observatory (NAOC) Scholarship**

2021 **Tsinghua Xuetang Scholarship**

2019-2021 **National Inspirational Scholarship** (3 Years)

Research Grant & Funding

2024–2025 **NASA Keck PI Data Awards**, USD 18,650 (US Admin X. Fan)

Awarded Telescope Proposals

PI or Contact **N = 14**

14 **VLA**, *26A-151*, PI Mingyu Li, 80 hrs
Mapping Widespread Molecular Gas across the Densest Protocluster of Lya Nebulae

13 **ALMA**, *2025.1.01223.S*, PI Haibin Zhang, 14.7 hrs
JAKS: JWST-ALMA-Keck Synergy Study on the Circumgalactic Cold Gas Accretion

[1] One-time award to the best senior doctoral student.
[2] The highest award to junior graduate student.

12 **Keck/KCWI**, *2025A N124*, PI Xiaohui Fan, 1.5 nights
JWST-ALMA-Keck Synergy Study on the Circumgalactic Gas Accretion in a Giant Lya Blob

11 **Magellan/LLAMAS**, *2025A*, PI Xiaojing Lin, 1.5 nights
Emission Line Mapping for Circumgalactic Cold Mode Accretion with LLAMAS Integral Field Spectroscopy

10 **CFHT/MegaCam**, *25AS13*, PI Mingyu Li, 1.5 nights
Probing Neutral Hydrogen Densities in The Most Massive Large Scale Structures at z=2

9 **MMT/Binospec**, *2024B*, PI Xiaojing Lin, 1.5 nights
Emission Line Mapping for Circumgalactic Cold Mode Accretion with MMT/Binospec Integral Field Spectroscopy

8 **Magellan/IMACS**, *2024B*, PI Zheng Cai, 1 night
The Cosmic Himalayas: Unveiling Interplay between Galaxies, QSOs, and IGM in the Densest Region at Cosmic Noon

7 **ALMA**, *2024.1.01259.S*, PI Haibin Zhang, 14.7 hrs
JAKS: JWST-ALMA-Keck Synergy Study on the Circumgalactic Cold Gas Accretion

6 **CFHT/MegaCam**, *24BS07*, PI Mingyu Li, 1 nights
Probing Neutral Hydrogen Densities in The Most Massive Large Scale Structures at z=2

5 **HST**, *Cycle 31 GO-17627*, 2 orbits
Origin and Feedback of a Giant Radio Galaxy in an Extreme Protocluster at z=2.3

4 **VLA**, *24B-358*, PI Mingyu Li, 12 hrs
Origin and Feedback of a Giant Radio Galaxy in an Extreme Protocluster at z=2.3

3 **Subaru/MOIRCS+HSC**, *S24A-091*, PI Yongming Liang, 2 nights + 3 hrs
Origin and properties of an intergalactic-scale, metal-enriched filament at z=2.3

2 **JCMT/SCUBA-2**, *M24AP004*, PI Yongming Liang, 8.7 hrs
Missing Power of the Largest Group of Enormous Lyman-alpha Nebulae at Cosmic Noon

1 **MMT/Binospec**, *2024A*, PI Yunjing Wu, 1 night
MMT/Binospec Spectroscopy of the Densest Group of Giant Lyα Blobs at Cosmic Noon

As CoI *(Selected Highlights)*

30 **JWST**, *Cyele 4 GO-7390*, PIs Yunjing Wu, Zheng Cai & Fengwu Sun, 26.2 hrs
Probing Pair-Instability Supernovae through the Triply-lensed MACS0647-JD at z=10.17

29 **JWST**, *Cyele 4 GO-7345*, PIs Yoshinobu Fudamoto & Fengwu Sun, 35.9 hrs
The Dragon survey: A Direct Probe of the Early Stellar Luminosity Function and Dark Matter through Multi-cycle Multi-cadence Microlensing at z=0.73

28 **JWST**, *Cyele 4 GO-7240*, PIs Bjorn Emonts, Pierre Guillard & Matthew Lehnert, 36.3 hrs
Gas cooling in the molecular CGM of the Spiderweb: in-situ galaxy growth at z=2

27 **VLA**, *26A-174*, PI Bjorn Emonts, 42.5 hrs
Prebiotic molecules in the Early Universe: the unique z=3.4 mm absorber B2 0902

26 **Chandra**, *Cyele 27 GO-27700187*, PI Yongming Liang, 340 ks
X-ray Census of Cosmic Himalayas: Super Massive Black Hole Growth at the Cosmic Extremes

25 **ALMA**, *2025.1.01615.S*, PI Yongming Liang
The Molecular Backbone of the Cosmic Himalayas: Mosiac CO(3-2) Mapping at z$\sim$2.2 with ALMA Band 3

24 **ALMA**, *2025.1.01243.S*, PI Yongming Liang
Cosmic Himalayas: Mapping Gas and Dust in the Hosts of the Densest QSOs at z$\sim$2 with ALMA Bands 4 and 7

23 **ALMA**, *2025.1.01077.S*, PI Tomokazu Kiyota
ALMA CO Mapping of the Highest-Density QSO Peak Identified Across a 10,000 deg^2 Sky

22 **ALMA**, *2025.1.00603.S*, PI Ben Wang
Probing the Molecular Circum-Galactic Medium Around 3 Hyper-Luminous Obscured QSOs with ALMA+ACA

21 **ALMA**, *2025.1.00215.S*, PI Fengwu Sun
Dust-Obscured Starbursts in the Most Massive Dark Matter Halos at z=8-9: an ALMA+JWST Wide-Field Census

20 **Chandra**, *Cyele 26 GO-26700289*, PI Yuanyuan Su, 120 ks
X-ray nuclear activity in an extreme protocluster with the densest population of Lyman alpha nebulae

19 **VLT**/**ERIS**, *114.27LA*, PI Alice Concas, 51 hrs
GOLDEN - Gas Outflows in Lensed Dwarfs with ERIS and NIRISS

18 **Subaru**/**MOIRCS**, *S24B0135N*, PI Yongming Liang, 2 nights
Interplay between the Most Extremely Concentrated Quasars and Galaxies

17 **JWST**, *Cycle 3 GO-5221*, PIs Maria Pudoka & Feige Wang, 12.9 hrs
Mapping Cosmic Structure Evolution: Characterizing Two Massive Galaxy Protoclusters Anchored by z>7.5 Luminous Quasars

16 **ALMA**, *2024.1.01241.S*, PI Ben Wang
Probing the Cold Molecular Circum-Galactic Medium Around Most Luminous Type-2 QSOs at z$\sim$2 with ALMA+ACA

15 **CFHT/MegaCam**, *24AS07*, PI Haibin Zhang, 1 night
Probing Neutral Hydrogen Densities in The Most Massive Large Scale Structures at z=2

14 **VLT/ERIS**, *113.26SV*, PI Alice Concas, 73 hrs
Probing High-Redshift Dwarf Galaxies and Their Gas Outflows with ERIS

13 **Subaru/MOIRCS**, *S24A0057N*, PI Rhythm Shimakawa, 1 night
Emission-line Mapping in a Protocluster at z=3 Fed by Inspiraling Cold Streams

12 **Magellan/IMACS+MagE**, *2024A*, PI Zheng Cai, 1 night
Exploring the most metal-poor DLA at cosmic noon

11 **Subaru/MOIRCS**, *S23B0117N*, PI Yongming Liang, 1 night
Interplay between the Most Extremely Concentrated Quasars and Galaxies

10 **JCMT/SCUBA-2**, *M23BP028*, PI Yongming Liang, 12.3 hrs
Interplay between Galaxies and the Most Extremely Concentrated Quasars

9 **VLT/MUSE**, *112.262L*, PI Emanuele Paolo Farina, 66 hrs
The cosmic Ecosystem of the first QSOs and Galaxies: a MUSE/XSHOOTER/JWST/ALMA Legacy Survey

8 **JWST**, *Cycle 2 GO-3325*, PI Feige Wang, 44.51 hrs + 16.34 hrs in parallel
Mapping the Most Extreme Protoclusters in the Epoch of Reionization

7 **JWST**, *Cycle 2 GO-2883*, PI Fengwu Sun, 38.73 hrs + 19.11 hrs in parallel
MAGNIF: Medium-band Astrophysics with the Grism of NIRCam in Frontier Fields

6 **ALMA**, *2023.1.01213.S*, PI Yunjing Wu
Revealing the dominant process that regulates gas-phase metallicity during the ongoing mergers at z$>$6

5 **VLT/MUSE**, *111.24UJ*, PI Fuyan Bian, 8 hrs
Filler Program for the Apocalypse: end all weather-idle-time on UT4 in P111

4 **Keck/DEIMOS**, *2022A, U030*, PI Alexie Leauthaud, 2 nights
Spectroscopic Follow-up of the Merian Survey: From Dark Matter in Dwarf Galaxies at z=0.1 to Proto Clusters at z=3-5

3 **ALMA**, *2022.1.01705.S*, PI Bjorn Emonts, 22.8 hrs
Physical properties of the molecular medium across the circum-galactic environment in a sample of four quasars

2 **Magellan/IMACS**, *2022B*, PI Qiong Li, 2 nights
Magellan/IMACS Multi-Slit Spectroscopy Observations for Lyα Emitters at z=2

1 **ALMA**, *2021.1.01142.S*, PI Bjorn Emonts, 45.5 hrs
The cold molecular medium across Enormous Ly-alpha Nebulae: studying clustered galaxy formation with ALMA+ACA

Leading-Author Publications

6 **Sijia Cai, Mingyu Li, Zheng Cai, et al.** 2025, *A Metal-free Galaxy at $z = 3.19$? Evidence of Late Population III Star Formation at Cosmic Noon*, ApJL, 993, L52

5 **Pengjun Lu, Mingyu Li, Dalya Baron, et al.** 2025, *Oxyster: A Circumgalactic Low-ionization Oxygen Nebula next to a Starburst Galaxy at $z \sim 1$*, ApJ, 990, 168

4 **Mingyu Li, Bjorn H. C. Emonts, Zheng Cai, et al.** 2025, *The Cosmic Owl: Twin Active Collisional Ring Galaxies with Starburst Merging Front at $z = 1.14$*, arXiv, arXiv:2506.10058

3 **Haibin Zhang, Zheng Cai, Mingyu Li, et al.** 2025, *MAMMOTH-Subaru. IV. Large Scale Structure and Clustering Analysis of Lyα Emitters and Lyα Blobs at $z = 2.2$–2.3*, ApJ, 981, 70

2 **Mingyu Li, Haibin Zhang, Zheng Cai, et al.** 2024, *MAMMOTH-Subaru. II. Diverse Populations of Circumgalactic Lyα Nebulae at Cosmic Noon*, ApJS, 275, 27

1 **Mingyu Li, Zheng Cai, Fuyan Bian, et al.** 2023, *The Mass-Metallicity Relation of Dwarf Galaxies at Cosmic Noon from JWST Observations*, ApJL, 955, L18

Conferences, Colloquia & Seminar Talks

November 2025 **Chinese Astronomical Society Annual Meeting 2025 (CAS2025)**, Xiamen, China, *Exploring the High-z Circumgalactic Meidum using IFU on GTC*

September 2025 **THU-DoA Doctoral Student Forum**, Beijing, China, *Birth, Breath, and Death of Galaxies: Direct Imaging of the High-z Galaxies and their Circumgalactic Medium*

July 2025 **Chinese Space Station Survey Telescope (CSST) Annual Conference**, Kunming, China, *Unveiling Extended and Extreme Line-emitting Gas in the Early Universe*

July 2024 **European Astronomical Society Annual Meeting 2024 (EAS2024)**, Padova, Italy, *Giant Lyα Blobs: Unveiling the Circumgalactic Medium of SMGs from Subaru, Keck, ALMA, and JWST*

May 2024 **THU-DoA Textbook Club** , Beijing, China, *Cool Gas in High-Redshift Galaxies*

April 2024 **Special Seminar in ICRR/NAOJ**, Tokyo, Japan, *Mapping the CGM emission at cosmic noon with multiwavelength observations*

November 2023 **THU-DoA Textbook Club**, Beijing, China, *The Circumgalactic Medium*

October 2023 **EREBUS Meeting 2023**, Biosphere-2, Arizona, USA, *Helium Emission at High-z Universe*

Teaching & Outreach

2023 **Teaching assistance of Observational Astronomy (30920023-0)** an undergraduate course given by Prof. Song Huang

2020-2021 **Director of Tsinghua Astronomy Observatory (0.4m)**, Tsinghua (Student) Astronomy Society. *Host amateur telescope instruction and monthly campus stargazing*

Academic References

Zheng Cai — Department of Astronomy, Tsinghua University (THU)
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Bjorn Emonts — National Radio Astronomy Observatory (NRAO)
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Fuyan Bian — European Southern Observatory (ESO)
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Xiaohui Fan — Steward Observatory, University of Arizona
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Full Publication List

Also see full list of publication on NASA/ADS

43 **Sijia Cai, Mingyu Li, Zheng Cai, et al.** 2025, *A Metal-free Galaxy at $z = 3.19$? Evidence of Late Population III Star Formation at Cosmic Noon*, ApJL, 993, L52

42 **Shany Danieli, Erin Kado-Fong, Song Huang, et al.** 2025, *First Data Release of the Merian Survey: A Wide-field Imaging Survey of Dwarf Galaxies at $z \sim 0.06$–0.10*, ApJ, 993, 110

41 **Zihao Li, Koki Kakiichi, Lise Christensen, et al.** 2025, *Insights on metal enrichment and environmental effects at $z \approx 5$–7 with JWST ASPIRE/EIGER and the chemical evolution model*, A&A, 703, A106

40 **Romain A. Meyer, Feige Wang, Koki Kakiichi, et al.** 2025, *JWST COSMOS-3D: Spectroscopic Census and Luminosity Function of [O III] Emitters at 6.75<z<9.05 in COSMOS*, arXiv, arXiv:2510.11373

39 **Zihao Li, Zheng Cai, Xin Wang, et al.** 2025, *A 13 Billion Year View of Galaxy Growth: Metallicity Gradient Evolution from the Local Universe to $z = 9$ with JWST and Archival Surveys*, ApJS, 280, 62

38 **Weizhe Liu, Xiaohui Fan, Huan Li, et al.** 2025, *Frequent Extreme Galaxy-scale Outflows among Luminous Early Quasars*, arXiv, arXiv:2509.08793

37 **Pengjun Lu, Mingyu Li, Dalya Baron, et al.** 2025, *Oxyster: A Circumgalactic Low-ionization Oxygen Nebula next to a Starburst Galaxy at $z \sim 1$*, ApJ, 990, 168

36 **Sijia Cai, Zheng Cai, Jianwei Lyu, et al.** 2025, *Discovery of a Pair of Galaxies with Both Hosting X-Ray Binary Candidates at $z = 2.544$*, ApJL, 989, L35

35 **Xiaojing Lin, Xiaohui Fan, Zheng Cai, et al.** 2025, *The Discovery of Little Red Dots in the Local Universe: Signatures of Cool Gas Envelopes*, arXiv, arXiv:2507.10659

34 **Maria Pudoka, Feige Wang, Xiaohui Fan, et al.** 2025, *Lyman-break Galaxies in the Megaparsec-scale Environments around Three $z \sim 7.5$ Quasars with JWST Imaging*, ApJ, 987, 198

33 **Shuqi Fu, Fengwu Sun, Linhua Jiang, et al.** 2025, *Medium-band Astrophysics with the Grism of NIRCam In Frontier Fields (MAGNIF): Spectroscopic Census of Hα Luminosity Functions and Cosmic Star Formation at $z \sim 4.5$ and 6.3*, ApJ, 987, 186

32 **Chenrui Ma, Zechang Sun, Tao Jing, et al.** 2025, *Can AI Dream of Unseen Galaxies? Conditional Diffusion Model for Galaxy Morphology Augmentation*, arXiv, arXiv:2506.16233

31 **Mingyu Li, Bjorn H. C. Emonts, Zheng Cai, et al.** 2025, *The Cosmic Owl: Twin Active Collisional Ring Galaxies with Starburst Merging Front at $z = 1.14$*, arXiv, arXiv:2506.10058

30 **Fengwu Sun, Jinyi Yang, Feige Wang, et al.** 2025, *The Identification of Two JWST/NIRCam-Dark Starburst Galaxies at $z = 6.6$ with ALMA*, arXiv, arXiv:2506.06418

29 **Yongming Liang, Masami Ouchi, Dongsheng Sun, et al.** 2025, *Cosmic Himalayas: The Highest Quasar Density Peak Identified in a 10,000 deg² Sky with Spatial Discrepancies between Galaxies, Quasars, and IGM H I*, ApJ, 986, 60

28 **Leyao Wei, Song Huang, Jiaxuan Li, et al.** 2025, *Zangetsu: A Candidate of Isolated, Quiescent, and Backsplash Ultra-Diffuse Galaxy in the COSMOS Field*, arXiv, arXiv:2505.14073

27 **Xiaojing Lin, Xiaohui Fan, Feige Wang, et al.** 2025, *Bridging Quasars and Little Red Dots: Insights into Broad-Line AGNs at $z = 5 - 8$ from the First JWST COSMOS-3D Dataset*, arXiv, arXiv:2504.08039

26 **Yoshinobu Fudamoto, Fengwu Sun, Jose M. Diego, et al.** 2025, *Identification of more than 40 gravitationally magnified stars in a galaxy at redshift 0.725*, NatAs, 9, 428

25 **Jaclyn B. Champagne, Feige Wang, Jinyi Yang, et al.** 2025, *A Quasar-anchored Protocluster at $z = 6.6$ in the ASPIRE Survey. II. An Environmental Analysis of Galaxy Properties in an Overdense Structure*, ApJ, 981, 114

24 **Jaclyn B. Champagne, Feige Wang, Haowen Zhang, et al.** 2025, *A Quasar-anchored Protocluster at $z = 6.6$ in the ASPIRE Survey. I. Properties of [O III] Emitters in a 10 Mpc Overdensity Structure*, ApJ, 981, 113

23 **Haibin Zhang, Zheng Cai, Mingyu Li, et al.** 2025, *MAMMOTH-Subaru. IV. Large Scale Structure and Clustering Analysis of Lyα Emitters and Lyα Blobs at $z = 2.2$–2.3*, ApJ, 981, 70

22 **Fengwu Sun, Feige Wang, Jinyi Yang, et al.** 2025, *A SPectroscopic Survey of Biased Halos in the Reionization Era (ASPIRE): Spectroscopically Complete Census of Obscured Cosmic Star Formation Rate Density at $z = 4$–6*, ApJ, 980, 12

21 **B. Peng, F. Arrigoni Battaia, A. Vishwas, et al.** 2025, *Direct high-resolution observation of feedback and chemical enrichment in the circumgalactic medium at redshift $z \sim 2.8$*, A&A, 694, L1

20 **Mingyu Li, Haibin Zhang, Zheng Cai, et al.** 2024, *MAMMOTH-Subaru. II. Diverse Populations of Circumgalactic Lyα Nebulae at Cosmic Noon*, ApJS, 275, 27

19 **Jianan Li, Bjorn H. C. Emonts, Zheng Cai, et al.** 2024, *The SUPERCOLD-CGM Survey. II. [C I](1–0) Emission and the Physical Conditions of Cold Gas in Enormous Lyα Nebulae at $z \sim 2$*, ApJ, 977, 198

18 **Xiangyu Jin, Jinyi Yang, Xiaohui Fan, et al.** 2024, *A SPectroscopic Survey of Biased Halos In the Reionization Era (ASPIRE): JWST Supports Earlier Reionization around [O III] Emitters*, ApJ, 976, 93

17 **Weizhe Liu, Xiaohui Fan, Jinyi Yang, et al.** 2024, *Fast Outflow in the Host Galaxy of the Luminous $z = 7.5$ Quasar J1007+2115*, ApJ, 976, 33

16 **Abby Mintz, Jenny E. Greene, Erin Kado-Fong, et al.** 2024, *A Non-parametric Morphological Analysis of Hα Emission in Bright Dwarfs Using the Merian Survey*, ApJ, 974, 273

15 **Xiaojing Lin, Feige Wang, Xiaohui Fan, et al.** 2024, *A SPectroscopic Survey of Biased Halos In the Reionization Era (ASPIRE): Broad-line AGN at $z = 4$-5 Revealed by JWST/NIRCam WFSS*, ApJ, 974, 147

14 **Yifei Luo, Alexie Leauthaud, Jenny Greene, et al.** 2024, *The Merian survey: design, construction, and characterization of a filter set optimized to find dwarf galaxies and measure their dark matter halo properties with weak lensing*, MNRAS, 530, 4988

13 **Xiaojing Lin, Zheng Cai, Yunjing Wu, et al.** 2024, *Quantifying the Escape of Lyα at $z \approx 5$–6: A Census of Lyα Escape Fraction with Hα-emitting Galaxies Spectroscopically Confirmed by JWST and VLT/MUSE*, ApJS, 272, 33

12 **Yuanhang Ning, Zheng Cai, Xiaojing Lin, et al.** 2024, *Unveiling Luminous Lyα Emitters at $z \approx 6$ through JWST/NIRCam Imaging in the COSMOS Field*, ApJL, 963, L38

11 **Siwei Zou, Zheng Cai, Feige Wang, et al.** 2024, *A SPectroscopic survey of biased halos In the Reionization Era (ASPIRE): Impact of Galaxies on the Circumgalactic Medium Metal Enrichment at $z > 6$ Using the JWST and VLT*, ApJL, 963, L28

10 **Ke Ma, Haibin Zhang, Zheng Cai, et al.** 2024, *MAMMOTH-Subaru. V. Effects of Cosmic Variance on Lyα Luminosity Functions at $z = 2.2$-2.3*, ApJ, 961, 102

9 **Haibin Zhang, Zheng Cai, Yongming Liang, et al.** 2024, *MAMMOTH-Subaru. III. Lyα Halo Identified by Stacking $\sim$3300 Lyα Emitters at $z = 2.2$–2.3*, ApJ, 961, 63

8 **Zihao Li, Zheng Cai, Fengwu Sun, et al.** 2023, *MAGNIF: A Tentative Lensed Rotating Disk at $z = 8.34$ detected by JWST NIRCam WFSS with Dynamical Forward Modeling*, arXiv, arXiv:2310.09327

7 **Yunjing Wu, Feige Wang, Zheng Cai, et al.** 2023, *A SPectroscopic Survey of Biased Halos in the Reionization Era (ASPIRE): JWST Discovers an Overdensity around a Metal Absorption-selected Galaxy at z 5.5*, ApJL, 956, L40

6 **Mingyu Li, Zheng Cai, Fuyan Bian, et al.** 2023, *The Mass-Metallicity Relation of Dwarf Galaxies at Cosmic Noon from JWST Observations*, ApJL, 955, L18

5 **Shiwu Zhang, Zheng Cai, Dandan Xu, et al.** 2023, *Revealing the Gas Recycling in the Circumgalactic Medium (CGM) Utilizing a Luminous Lya Nebula around a Type II Quasar at z = 2.6 with the Keck Cosmic Web Imager (KCWI)*, ApJ, 952, 124

4 **Feige Wang, Jinyi Yang, Joseph F. Hennawi, et al.** 2023, *A SPectroscopic Survey of Biased Halos in the Reionization Era (ASPIRE): JWST Reveals a Filamentary Structure around a z = 6.61 Quasar*, ApJL, 951, L4

3 **Xiaojing Lin, Zheng Cai, Siwei Zou, et al.** 2023, *Metal-enriched Neutral Gas Reservoir around a Strongly Lensed Low-mass Galaxy at z = 4 Identified by JWST/NIRISS and VLT/MUSE*, ApJL, 944, L59

2 **Yunjing Wu, Zheng Cai, Fengwu Sun, et al.** 2023, *The Identification of a Dusty Multiarm Spiral Galaxy at z = 3.06 with JWST and ALMA*, ApJL, 942, L1

1 **Jianrui Li, Bjorn H. C. Emonts, Zheng Cai, et al.** 2021, *Massive Molecular Outflow and 100 kpc Extended Cold Halo Gas in the Enormous Lyα Nebula of QSO 1228+3128*, ApJL, 922, L29